Experiment 10: End-to-End MLOps Pipeline Implementation

AIM

To implement an End-to-End MLOps pipeline integrating CI/CD, versioning, model deployment, and comparison between AutoML and traditional ML.

OBJECTIVES

1. To build a complete ML workflow from training to deployment.
2. To compare traditional ML models with AutoML pipelines.
3. To implement experiment tracking and CI/CD automation.

THEORY

1. MLOps Concept

MLOps combines Machine Learning with DevOps practices to automate training, testing, deployment, and monitoring. It ensures reproducibility, scalability, and structured ML lifecycle management.

2. Traditional ML vs AutoML

Traditional ML requires manual feature selection and tuning.
AutoML tools like TPOT and Auto-sklearn automatically generate optimized pipelines, reducing manual effort and speeding up experimentation.

3. Experiment Tracking & Versioning

MLflow is used to log parameters, metrics, and models. It maintains version control and supports model registry for structured management.

4. CI/CD and Deployment

CI/CD automates retraining and deployment when code changes occur. Models are containerized using Docker and deployed locally (offline) or to cloud platforms like Azure Machine Learning or Google Cloud Vertex AI.

PROCEDURE

1. Preprocess dataset using Pandas and split into train/test sets.
2. Train traditional models (e.g., Logistic Regression, Random Forest).
3. Generate optimized pipelines using TPOT/AutoSklearn.
4. Compare accuracy using Matplotlib and Seaborn visualizations.
5. Track experiments using MLflow.
6. Automate training with CI/CD.
7. Deploy model locally and to cloud endpoints.

OFFLINE SETUP

- Local data storage
- MLflow local tracking
- Docker containerization
- Local CI testing

ONLINE SETUP

- Azure ML / Vertex AI pipeline
- Cloud model registry
- Managed online endpoints
- Automated CI/CD

CONCLUSION

The experiment demonstrated an End-to-End MLOps pipeline integrating training, AutoML comparison, experiment tracking, CI/CD automation, and deployment. AutoML speeds up model optimization, while traditional ML offers better control.

Proper MLOps implementation ensures scalable, reproducible, and production-ready ML systems.